

Verification

o4_H_Simulator_FVM_1D_Compressible_Heterogeneous

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium.

2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x} \left(\frac{\rho k(x)}{\mu} \frac{\partial P}{\partial x} \right), \quad (1)$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m³],
- ϕ is porosity [-],
- $k(x)$ is permeability [m²],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}, \quad (2)$$

and assuming constant porosity and compressibility, the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \quad (3)$$

where ρ_0 is the reference density at pressure P_0 . For small pressure variations, the density may be linearized as

$$\rho \approx \rho_0 [1 + c_f(P - P_0)], \quad (4)$$

which leads to the pressure diffusivity equation,

$$\phi c_f \frac{\partial P}{\partial t} = \frac{\partial}{\partial x} \left(\frac{k(x)}{\mu} \frac{\partial P}{\partial x} \right). \quad (5)$$

The domain is defined on

$$0 \leq x \leq L_x, \quad (6)$$

with fixed-pressure boundary conditions

$$P(0, t) = P_L, \quad P(L_x, t) = P_R, \quad (7)$$

and the initial condition

$$P(x, 0) = P_i. \quad (8)$$

3 Analytical Solution

To validate the numerical simulator, a steady-state solution is used. In steady state, the flux is constant such that,

$$\frac{d}{dx} \left(k \frac{dP}{dx} \right) = 0. \quad (9)$$

As the flux is constant everywhere, it follows that the Darcy fluxes of the two sections of permeability are equivalent,

$$q = -\frac{k}{\mu} \frac{dP}{dx}. \quad (10)$$

The pressure gradients are therefore different and equal to

$$\frac{dP}{dx} = -\frac{\mu q}{k} \quad (11)$$

in each section. In the first layer, therefore,

$$P(x < a) = P_L - \frac{\mu q}{k_1} x \quad (12)$$

and in the second layer,

$$P(x > a) = P(a) - \frac{\mu q}{k_2} (x - a) \quad (13)$$

where a is the permeability change location with,

$$P(a) = P_L - \frac{\mu q}{k_1} a. \quad (14)$$

Then, solving for q yields,

$$q = \frac{P_L - P_R}{\mu \left(\frac{a}{k_1} + \frac{L_x - a}{k_2} \right)}. \quad (15)$$

By substituting in q ,

$$P(x) = \begin{cases} P_L - \frac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}} \frac{x}{k_1}, & 0 \leq x \leq a, \\ P_L - \frac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}} \left(\frac{a}{k_1} + \frac{x - a}{k_2} \right), & a \leq x \leq L_x. \end{cases} \quad (16)$$

4 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Number of cells	N_x	10000
Domain length	L_x	10 m
First permeability domain	a	2 m
Permeability	k_1	$1 \times 10^{-13} \text{ m}^2$
Permeability	k_2	$1 \times 10^{-12} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Left pressure	P_L	$10 \times 10^6 \text{ Pa}$
Right pressure	P_R	$1 \times 10^5 \text{ Pa}$
Compressibility	c_f	$1 \times 10^{-8} \text{ Pa}^{-1}$
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$

Table 1: Simulation parameters.

5 Results

5.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

6 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution at steady state.

7 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional slightly compressible flow in a heterogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

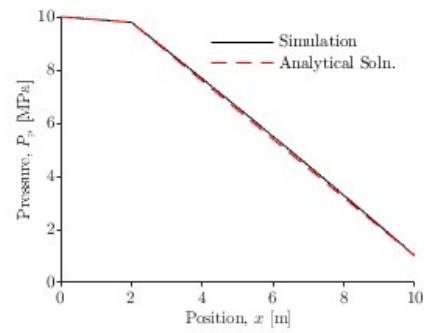


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

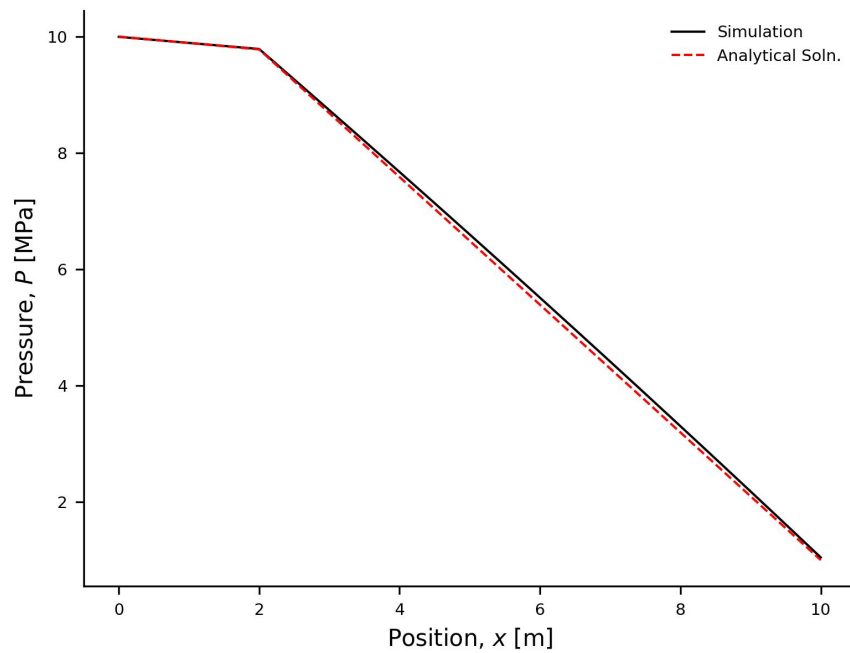


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.